

Sriram Seelamneni

github.com/sriramsme | srirams.me | linkedin.com/in/sriramsme

Professional Experience

Full-stack Software Engineer

Airlink Flight School | 2025 - current

- Primary engineer on a 4-person team that built Airlink's web and mobile management platform from scratch — admissions, scheduling, grading, lesson planning, and reporting for 75+ users across several roles.
- Solved offline-first **data sync** for in-flight grading: instructors score live metrics mid-air on unreliable connections, so built local persistence with conflict-aware sync on reconnect to keep training records consistent across devices.
- Designed secure, role-based workflows using **React, TypeScript, React Native, and Firebase Auth**, supporting students, instructors, and administrators under strict permission boundaries.
- Built backend services with Python/FastAPI on Firestore for real-time scheduling with automatic conflict detection, replacing a group chat coordination system, and moved admissions fully online, eliminating back-and-forth email chains.
- Deployed containerized services on **Google Cloud Platform (Docker)**, automating workflows that replaced multiple manual workflows end-to-end — in-flight scoring, admissions, and FAA-required grade sheet generation; administrative workload down ~60–70% across the board.

Product Engineer

trulyKin | 2025

- Built and shipped trulyKin solo, a private group-based family social app (**React Native, Node.js, Supabase/PostgreSQL**) covering posts, events, and real-time network management, from system design through App Store deployment.
- Designed and implemented a **dynamic family tree engine** from scratch — graph-based data model with members as nodes on an interactive canvas, edge traversal surfacing **2nd and 3rd degree** relationships across connected family networks.
- Built fine-grained privacy controls, real-time event subscriptions, and cross-family connection handling with role-scoped data access across the full backend.
- Reached 35+ organic users within the first months of launch with zero marketing spend.

Software Engineer

One Community Global | 2024-2025

- Contributed as a MERN (React.js/MongoDB) full-stack developer to build open-source Highest Good Network software.
- Built dashboard components and a supplier performance graph, fixed role-based permission leaks, and improved frontend load times by **20–30%** through caching improvements, redundant API call removal, and code optimization.
- Trained as a team manager, helped manage a team of 7, and raised PR issues to improve task allocation and progress tracking.
- Reviewed over **40 PRs & created 15+ PRs**, including developing new components, adding features, and fixing critical bugs.

Info-Tech Assistant

Computational Mathematics, Science & Eng dept, MSU | 2023-2024

- Automated departmental data management — replaced manual Excel-based workflows with a Python/SQLite pipeline aggregating data across multiple sources, reducing administrative overhead significantly.
- Built publication tracking and reporting tooling by integrating ORCID and Zotero APIs, automated annual metrics reports covering faculty research output, funding, and publication volume
- Facilitated the transition from an outdated format to a new, optimized structure, enhancing data accessibility and relevance for departmental use

Projects

OnlyAgents: Self-hosted autonomous AI agent framework | [github/OnlyAgents](https://github.com/OnlyAgents)

- Built an open-source multi-agent runtime in Go with a hierarchical executive–worker architecture for intent decomposition, delegation, and capability-bounded execution.
- Designed a kernel-mediated execution model (typed event bus, no direct agent-to-agent calls) to enforce isolation, observability, and strict trust/cost boundaries across agents.
- Implemented dynamic tool gating and planning phases (meta-tools vs execution tools) and runtime capability injection (AttachSkill + prompt rebuild), enabling structured reasoning and live system reconfiguration without restart.
- Engineered a 4-layer memory system (working memory, semantic episode search, knowledge graph with time-scoped entity-relation triples, and behavioral pattern extraction) with parallel retrieval assembled into agent context before each LLM call.
- Achieved highly optimized deployment footprint (36MB static binary, ~27MB RAM idle, 31ms startup), enabling portable, dependency-free execution vs. >1GB RAM / >500s startup in comparable frameworks.

Text-to-Action: Natural Language Interface Systems | [github/text-to-action](#)

- Built an open-source system translating natural language requests into executable application actions across APIs, chatbots, and voice-driven workflows.
- Architected semantic retrieval and parameter extraction pipelines using LLMs and NER; adopted over **1,500+ installs**.

Picko: Personalized Recommendation System | [picko.app](#)

- Developed and shipped a recommendation engine using ML techniques (TF-IDF, cosine similarity, semantic search) for personalized movies, TV shows, and books suggestions.
- Built and optimized a **300K+ item** relational dataset with high-concurrency query paths for real-time recommendations.
- Automated ingestion and refresh pipelines via CI-driven cron workflows, ensuring continuously updated recommendations.

aiso : AI-native social network | [laiso.com](#)

- Built an AI-native professional networking platform where an agent conducts user interviews via text message (Twilio), builds intent and career-stage profiles, and makes introductions to matched users directly in-thread — Python, Supabase, React, LLM-powered.
- Designed the matching pipeline around declared intent (co-founding, hiring, fundraising) cross-referenced against behavioral and career signals collected during onboarding.

PyOrcid : Open-source lib for Orcid Integration | [github/PyOrcid](#)

- Developed an open-source API client to help researchers and academic institutions to programmatically sync researcher profiles and publication data from ORCID.
- Automated access to public and authenticated datasets, reducing manual profile maintenance for downstream consumers.
- Achieved **1,000+ downloads** across academic and research communities.

Enhanced Video Assistant (EVA) - TechSmith | [eva/techsmith](#)

- Developed an award-winning production-grade video assistant app as a capstone project.
- Integrated machine learning models to automate audio enhancement, provide multi-layered feedback, and condense video to key moments, improving user experience and efficiency
- Managed the app deployment on Azure web services with Docker, for better scalability and reliability

Skills

- **Programming Languages:** Python, Go, C++, TypeScript
- **Cloud & Infrastructure:** Google Cloud Platform (GCP), Docker, CI/CD, Supabase, Cloudflare
- **AI/ML:** Agent frameworks, LLM Integration, PyTorch, scikit-learn, Prompt engineering, Embedding & Semantic Search
- **Backend & APIs:** FastAPI, Node.js, RESTful API design, OAuth
- **Data & Storage:** PostgreSQL, MongoDB, Firestore
- **Frontend & Mobile:** React, React Native, Expo, Tailwind CSS

Education

Michigan State University, College of Engineering | 2021-2024

Bachelor of Science, Computer Science

GPA: 3.84

Coursework: Computational Mathematics, Science & Engineering (CMSE) minor, Business cognate

Awards and Achievements

- **Auto-Owners Exposition Award** - Enhanced Video Assistant, April 2024
- **Dean's Scholarship** - \$75,000 merit-based scholarship, 2021-2024
- **Best Emerging Technology Award** - Brainwaive, January 2023